

Emotion Recognition from Photoplethysmography (PPG) Signals

Course: CSC 491/591 — Ubiquitous Computing and Mobile Health

Dataset: WESAD (Wearable Stress and Affect Detection)

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Abstract

In this course project, I developed a complete pipeline to recognize three emotional states : Baseline, Stress, and Amusement from wrist-worn PPG signals in the WESAD dataset [1]. I describe signal preprocessing, extract 11 physiological features grounded in heart rate variability (HRV) research, and train four classifiers spanning traditional machine learning and deep learning. Using a subject-level train/test split to prevent data leakage, the best model (1D CNN) achieves **64.8% accuracy** and **55.6% macro-F1** on three held-out subjects. For context, the original WESAD paper reported wrist BVP single-modality results ranging from approximately 64% to 70% accuracy depending on the classifier and evaluation protocol [1]; direct numerical comparisons with my results should be interpreted cautiously, as the two studies differ in window length (8 s vs. 60 s), evaluation design (3-subject holdout vs. full LOSO), and classifier set. All results are from actual model runs on the full 15-subject dataset.

Section 1 : Data Processing (60 pts)

1.1 Dataset Overview

The WESAD dataset [1] contains multimodal physiological recordings from 15 subjects (S2–S17, excluding S12, mean age 27.5 ± 2.4 years, 12 male and 3 female) collected during a controlled lab study. Each subject wore two devices: a RespiBAN Professional on the chest (700 Hz) and an Empatica E4 on the non-dominant wrist. This project uses only the wrist-worn BVP signal from the Empatica E4 (64 Hz), focusing on the wrist BVP single-modality condition, which is one of several evaluated in the original benchmark [1].

Study Protocol

The study was designed to reliably elicit three distinct affective states [1]:

- **Baseline (label 1):** 20-minute resting period in which subjects read neutral magazine content, inducing a calm, parasympathetically dominant state.
- **Stress (label 2):** Subjects underwent the Trier Social Stress Test (TSST), consisting of a 5-minute speech before a three-person evaluation panel followed by a 5-minute serial subtraction task (counting from 2023 in steps of 17). The social-evaluative design reliably induces sustained sympathetic activation [1].
- **Amusement (label 3):** Subjects watched 11 funny video clips (~392 seconds total), each followed by a 5-second neutral sequence to prevent carry-over.
- **Meditation (labels 4–7, excluded):** Guided breathing exercises placed after each stimulus condition; not used for classification.

To control for order effects, half the subjects experienced Amusement before Stress and the other half experienced Stress before Amusement. The wrist label array is encoded as integers: only labels 1 (Baseline), 2 (Stress), and 3 (Amusement) are used; all others are discarded. Subject S2 is used throughout Section 1 as a representative demonstration; the identical pipeline is applied to all 15 subjects in Section 2.

After the full pipeline, subject S2 yields **531 segments**: Baseline 286 (53.9%), Stress 154 (29.0%), Amusement 91 (17.1%). This distribution closely matches the proportions reported in the original WESAD paper (53%, 30%, 17%) [1], confirming that the pipeline correctly recovers the intended conditions.

1.2 Step 1 : Butterworth Bandpass Filtering

Raw wrist PPG signals contain three main noise sources: **baseline wander** (< 0.5 Hz drift from breathing and movement), **high-frequency noise** (> 4 Hz from electronics and ambient light), and **motion artifacts** from wrist displacement. A **4th-order Butterworth bandpass filter** (0.5–4.0 Hz) is applied via `scipy.signal.filtfilt` to remove these while preserving the cardiac frequency band (30–240 BPM). The Butterworth design is chosen for its maximally flat passband — it introduces no ripple that would distort PPG waveform morphology. Zero-phase filtering via `filtfilt` (forward-backward pass) preserves the precise timing of systolic peaks, which is essential for accurate IBI computation. The lower cutoff (0.5 Hz) removes baseline wander and respiration-induced drift; the upper cutoff (4.0 Hz) attenuates high-frequency motion noise while passing all physiologically plausible heart rates. The 4th-order rolloff (-80 dB/decade) is sharp enough to suppress out-of-band noise without introducing ringing.

Figure 1 shows the effect of filtering on the first five seconds of the S2 recording.

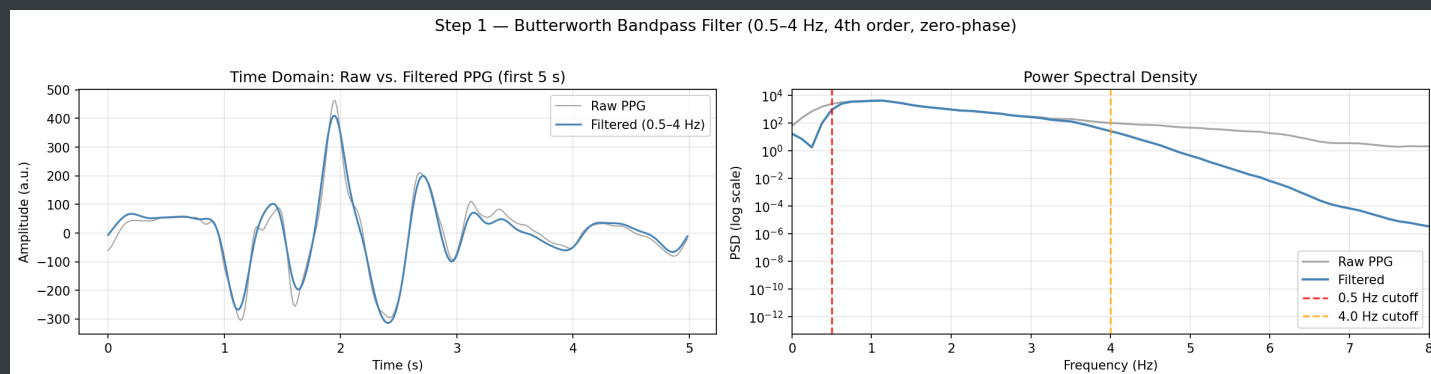


Figure 1. Left: Time-domain comparison of raw (gray) and filtered (blue) PPG over 5 seconds. Baseline drift is removed and cardiac pulses are clearly defined. Right: Power spectral density (log scale). Dashed lines mark the 0.5 Hz and 4.0 Hz cutoffs; out-of-band power is strongly attenuated.

1.3 Step 2 : Label Alignment

The PPG signal is sampled at 64 Hz and the label array at 700 Hz. Each PPG sample index i is mapped to its corresponding label sample index via $\text{floor}(i \times 700 / 64)$, clipped to valid bounds. This produces a per-sample label array aligned to the filtered PPG. Only samples with labels 1, 2, or 3 are retained.

1.4 Step 3 : Sliding Window Segmentation

The recording must be divided into fixed-size windows to provide unambiguous single-label inputs for supervised learning and to generate sufficient training samples from a small subject pool. I use **8-second windows (512 samples at 64 Hz)** with a **4-second step (50% overlap)**.

The choice of 8-second windows reflects a practical balance between sample density, waveform learning, and HRV validity. An 8-second window captures 8–15 cardiac cycles at typical resting and stress heart rates, which is the practical minimum for computing time-domain HRV metrics such as RMSSD and SDNN (at least 5 consecutive beats are required). Crucially, short windows produce substantially more training samples than long ones: an 8-second step yields approximately 550 segments per subject, compared to roughly 50–60 segments that a 60-second window would permit from the same recording. This denser sampling is especially important for end-to-end CNN training, where more windows per subject allow the model to learn waveform-level morphological patterns such as pulse shape, diastolic notch prominence, and systolic amplitude variation, rather than relying solely on derived statistics. The 50% overlap approximately doubles the number of usable windows. Each window is assigned the label covering the majority of its samples ($> 50\%$); windows spanning label transitions are discarded.

Window length trade-off: This design deliberately prioritizes sample density and waveform-level learning at the cost of spectral HRV resolution. Both the original WESAD benchmark [1, Section 4.1] and the hybrid CNN study [2] used 60-second windows (3,840 samples), which is the conventional choice for reliable LF and HF power estimation. Shorter 8-second windows are better suited to dense sample generation and CNN-based end-to-end learning, but the 0.125 Hz spectral resolution is coarser than the LF band width (~ 0.11 Hz), making frequency-domain metrics such as LF/HF approximate. This trade-off is discussed further in Section 1.5 and the

Limitations.

Figure 2 shows the segmentation over a 40-second window of S2's recording centered on the onset of the Stress condition.

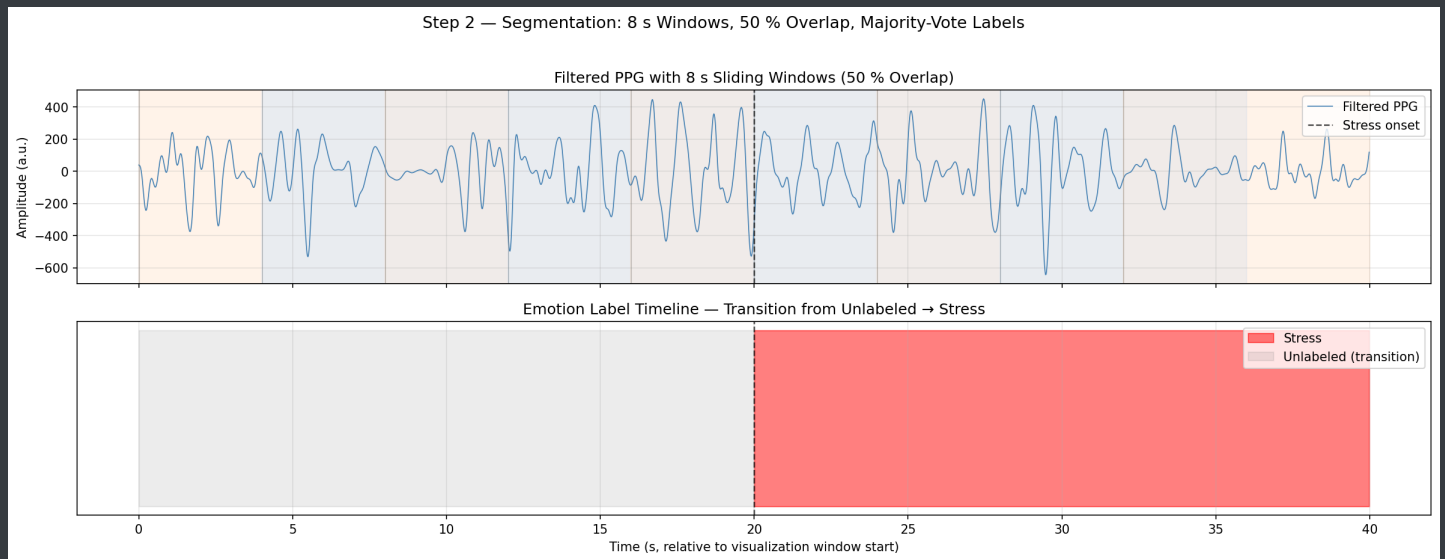


Figure 2. *Top: Filtered PPG with alternating shaded 8-second windows (50% overlap). The dashed line marks the onset of the Stress condition. Bottom: Emotion label timeline — the gray region is the unlabeled transition preceding Stress; the red region is Stress (label 2). Windows whose majority label is unlabeled are discarded.*

1.5 Step 4 : HRV Feature Extraction

I extract **11 features** per 8-second window. Systolic peaks are detected with `scipy.signal.find_peaks (distance = 32 samples , height = 0)`; IBIs are computed as $(\text{peak}[i+1] - \text{peak}[i]) / f_s$. Sympathetic activation during stress typically raises HR and suppresses beat-to-beat variability, while parasympathetic dominance at rest produces the opposite pattern [5].

PPG amplitude statistics:

- **mean:** average signal amplitude within the window
- **std:** amplitude variability

- **min**: minimum amplitude
- **max**: maximum amplitude
- **median**: robust central tendency

HRV-related features:

- **HR** = $60 / \text{mean}(\text{IBI})$: mean heart rate; typically elevated under stress
- **mean_IBI** = $\text{mean}(\text{IBI})$: average inter-beat interval; usually shorter under stress
- **SDNN** = $\text{std}(\text{IBI})$: overall beat-to-beat variability
- **RMSSD** = $\sqrt{\text{mean}(\text{diff}(\text{IBI})^2)}$: short-term parasympathetic HRV
- **pNN50**: percentage of successive IBI differences above 50 ms
- **LF/HF**: sympathovagal balance estimate (Welch PSD on 4 Hz IBI tachogram)

Note on LF/HF: An 8-second window gives spectral resolution of 0.125 Hz, which is coarser than the LF band width (~0.11 Hz). The LF/HF ratio is therefore an approximation; clinical standards recommend at least 5-minute windows [6]. The original WESAD benchmark used 60-second windows for HRV features [1, Section 4.1], as did Rashid et al. [2]. LF/HF is retained as a supplementary feature with this limitation acknowledged.

Figure 3 shows distributions of four representative HRV features for subject S2, illustrating the degree of class separation.

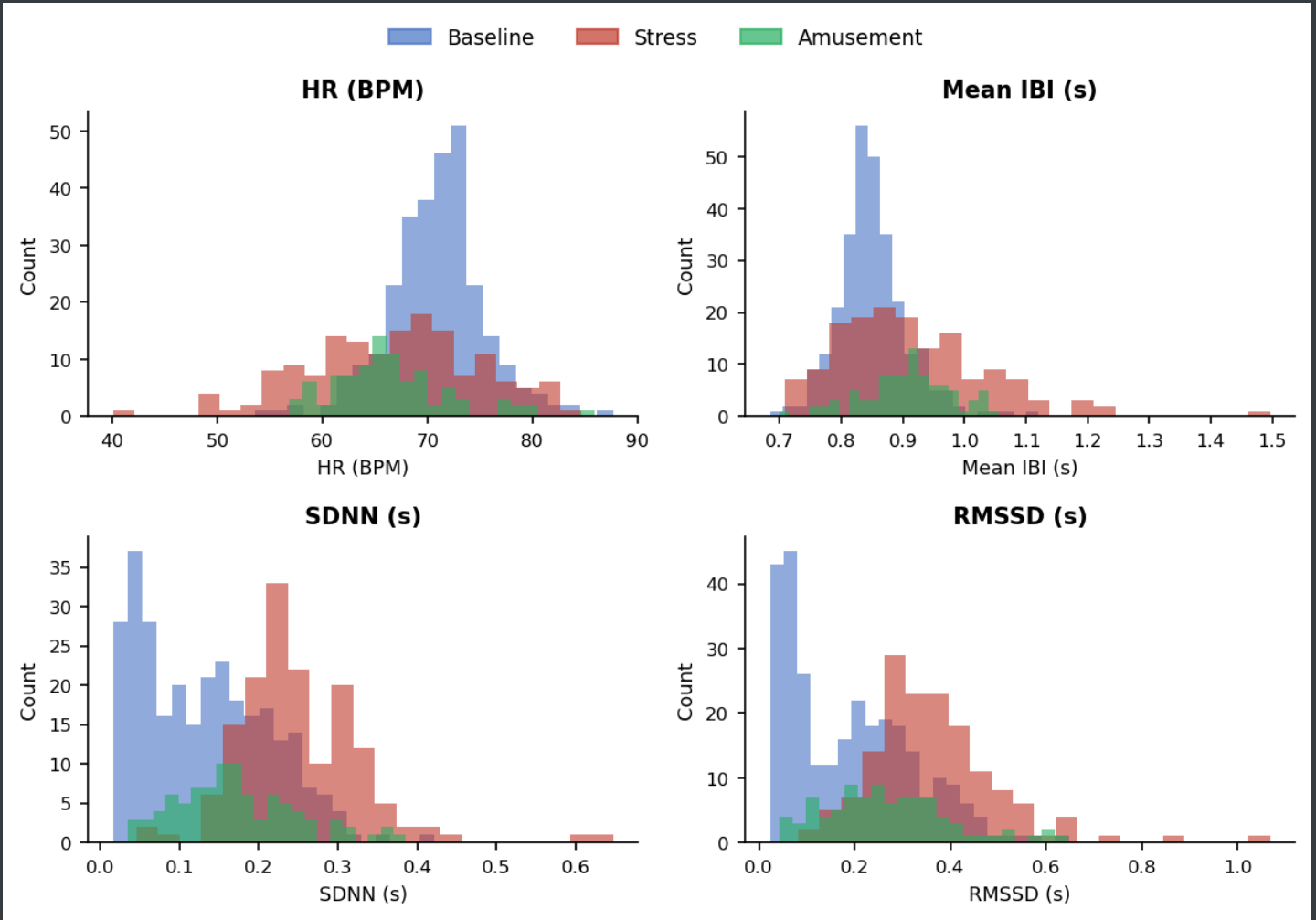


Figure 3. Distributions of four key HRV features for subject S2 by emotion class. HR and mean_IBI show clear Stress vs. Baseline separation consistent with sympathetic activation. SDNN and RMSSD show reduced variability under Stress. Baseline and Amusement overlap substantially across all features, foreshadowing the classification difficulty discussed in Section 3.

The pipeline produces a feature matrix X_{s2} of shape **(531, 11)** and a label vector y_{s2} of shape **(531,)**. Applied identically across all 15 subjects, it yields a combined dataset of 8,287 windows used in Section 2.

1.6 Literature Review

Two prior works are directly relevant to this project.

Schmidt et al. (2018) [1] introduced the WESAD dataset and established benchmark results using handcrafted physiological features and classical machine learning. The paper characterized multiple wrist and chest modalities, both individually and in combination, under Leave-One-Subject-Out (LOSO) cross-validation, making it the primary reference for the three-class Baseline / Stress / Amusement setting used here. For the wrist BVP single-modality condition, results ranged from approximately 64% to 70% accuracy depending on the classifier, with the wrist consistently underperforming the chest modality. The paper also documented the self-reported affective responses confirming that TSST reliably induced acute stress and that the physiological distinction between Amusement and Baseline was subtle, a finding central to the analysis in Section 3. The WESAD benchmark adopted 60-second sliding windows, reflecting the conventional requirement for reliable spectral HRV estimation.

Rashid et al. (2021) [2] focused specifically on wrist-based BVP / PPG as a single modality and proposed a hybrid convolutional neural network (H-CNN) that combines handcrafted HRV-related features, including mean and standard deviation of HR and HRV, pNN50, LF/HF, and frequency-band energy components, with representations automatically learned by a 1D CNN. The core argument is that feature-based and end-to-end approaches are complementary: handcrafted features encode established physiological domain knowledge, while CNN-learned features capture waveform morphology that is difficult to summarize with statistics alone. Evaluated via LOSO on WESAD with 60-second windows, the H-CNN achieved 75.21% accuracy and 64.15% macro-F1 for three-class classification, outperforming the best traditional classifier by approximately 5% in accuracy and 10% in macro-F1, and outperforming the normal CNN alone by approximately 7% in both metrics.

This project is related to both works. Like [1], I use the WESAD dataset, the same three-class label set, and a subject-independent evaluation design. Like [2], I restrict the analysis to the wrist BVP signal and compare feature-based classifiers against a 1D CNN. However, my implementation does not reproduce the full hybrid architecture of [2]: I train the CNN end-to-end on filtered PPG waveforms and train the classical models on handcrafted HRV features, but I do not concatenate the two feature streams into a single hybrid model. Rashid et al. [2] therefore represents a natural extension of this work, not a baseline I directly replicate.

Section 2 : Methodology (30 pts)

2.1 Multi-Subject Data Preparation

Applying the Section 1 pipeline to all 15 subjects produces the following segment counts:

Subject	Total	Baseline	Stress	Amusement
S2	531	286	154	91
S3	539	285	160	94
S4	541	289	159	93
S5	555	300	161	94
S6	550	295	162	93
S7	549	296	160	93
S8	552	292	167	93
S9	549	295	161	93
S10	569	295	181	93
S11	557	295	170	92
S13	557	295	166	96
S14	556	295	168	93
S15	559	294	172	93
S16	555	295	168	92
S17	568	295	180	93
Total	8,287	4,406	2,489	1,392

Combined dataset: `X_all` shape (8,287 × 11), `y_all` shape (8,287,)

Train / Test Split by Subject

Data is split by *subject*, not by *window*. Randomly splitting windows would allow the model to observe some windows from every subject's recording during training, effectively learning subject-specific physiological baselines rather than emotion-specific patterns, which constitutes a form of data leakage. This reflects the same underlying principle as the original WESAD benchmark, which adopted Leave-One-Subject-Out (LOSO) cross-validation to measure generalization to previously unseen individuals [1, Section 4.3]. Because physiological signals are highly person-dependent, any evaluation that does not hold out complete subjects overstates real-world performance.

Split	Subjects	Segments
Train	S2–S11, S13, S14 (12 subjects)	6,605
Test	S15, S16, S17 (3 subjects)	1,682

`StandardScaler` is fit on training data only and applied to both splits. All hyperparameter searches use `GroupKFold(n_splits=6)` with subject IDs as groups, so each CV fold holds out two complete subjects. Using `StratifiedKFold` instead inflates CV macro-F1 by ~4 percentage points (0.5308 vs. 0.4885), indicating a measurable effect of within-subject leakage.

I evaluate two complementary approaches: **feature-based classifiers** (Random Forest, Logistic Regression, SVM) trained on the 11 HRV features, and an **end-to-end 1D CNN** trained directly on filtered 512-sample PPG windows (per-segment normalized).

2.2 Model 1 : Random Forest

Random Forest trains an ensemble of decision trees on bootstrapped subsets of the data. At each split a random feature subset is evaluated, decorrelating the trees and reducing variance. Final predictions use majority vote. The model handles non-linear feature interactions naturally, requires no feature scaling, and provides feature importances. It is also among the classifiers evaluated in the original WESAD benchmark, where ensemble methods were among the better-performing approaches for wrist BVP classification [1].

Hyperparameters were tuned via GridSearchCV with subject-independent GroupKFold ($n_splits=6$, scoring: macro-F1), searching $n_estimators$ in $\{100, 200, 300\}$, max_depth in $\{None, 5, 10\}$, and $min_samples_leaf$ in $\{1, 2\}$. The best configuration ($n_estimators=300$, $max_depth=10$, $min_samples_leaf=1$) achieved a CV macro-F1 of **0.4885**.

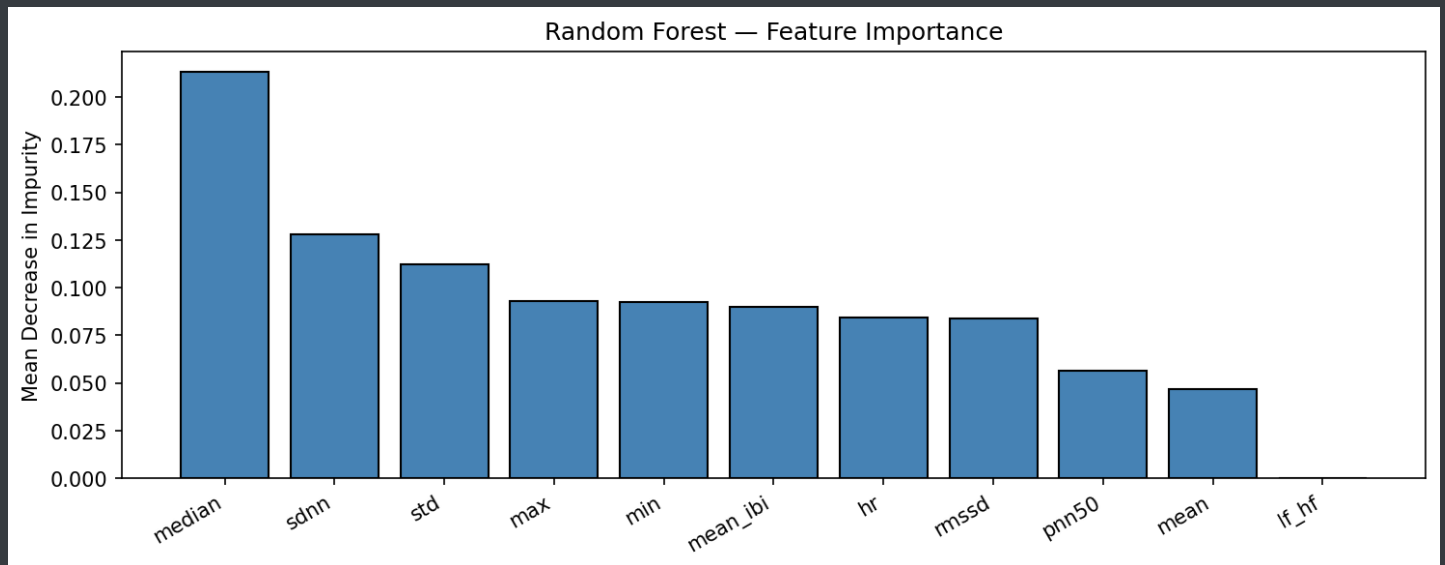


Figure 4. Feature importances from the tuned Random Forest. HRV time-domain metrics (RMSSD, SDNN, mean_IBI, HR) rank as most informative, consistent with the emphasis on HRV features in the WESAD benchmark and the hybrid CNN study [1, 2].

2.3 Model 2 : Logistic Regression

Logistic Regression finds a linear decision boundary via softmax, trained with L2 regularization and $max_iter=1000$ on standardized features; the regularization strength used the default scikit-learn setting ($C=1.0$). It serves as a linear baseline, assessing how much emotion signal is linearly separable in HRV feature space. Because the physiological distinction between Amusement and Baseline is subtle and likely non-linear, I expect this model to struggle, a pattern broadly consistent with the WESAD benchmark, where linear classifiers tended to underperform non-linear ensemble methods on the three-class task [1].

2.4 Model 3 : Support Vector Machine (RBF Kernel)

SVM with RBF kernel ($C=1.0$, $\gamma=\text{'scale'}$) maps scaled features into a high-dimensional space and finds the maximum-margin hyperplane. Feature scaling is essential: without it, high-range features dominate the kernel distance metric. $\gamma=\text{'scale'}$ sets $\gamma = 1/(\text{n_features} \times \text{Var}(X_{\text{train}}))$, automatically adapting to feature scale.

2.5 Model 4 : 1D Convolutional Neural Network

The 1D CNN operates directly on filtered PPG windows (512 samples), learning its own representations from waveform morphology without hand-crafted features. This is an end-to-end approach not included in the original WESAD benchmark, providing a direct comparison between domain-knowledge-driven and data-driven feature learning.

Input windows are normalized individually using per-segment z-score normalization: each 512-sample window is centered and scaled by its own mean and standard deviation ($(x - \mu_{\text{seg}}) / \sigma_{\text{seg}}$). This differs from the `StandardScaler` applied to the classical feature pipelines, which fits a single global mean and variance on the training set. Per-segment normalization removes inter-subject and inter-window amplitude baseline differences directly in the time domain, without reference to any global statistics. Figure 5a shows the full architecture.

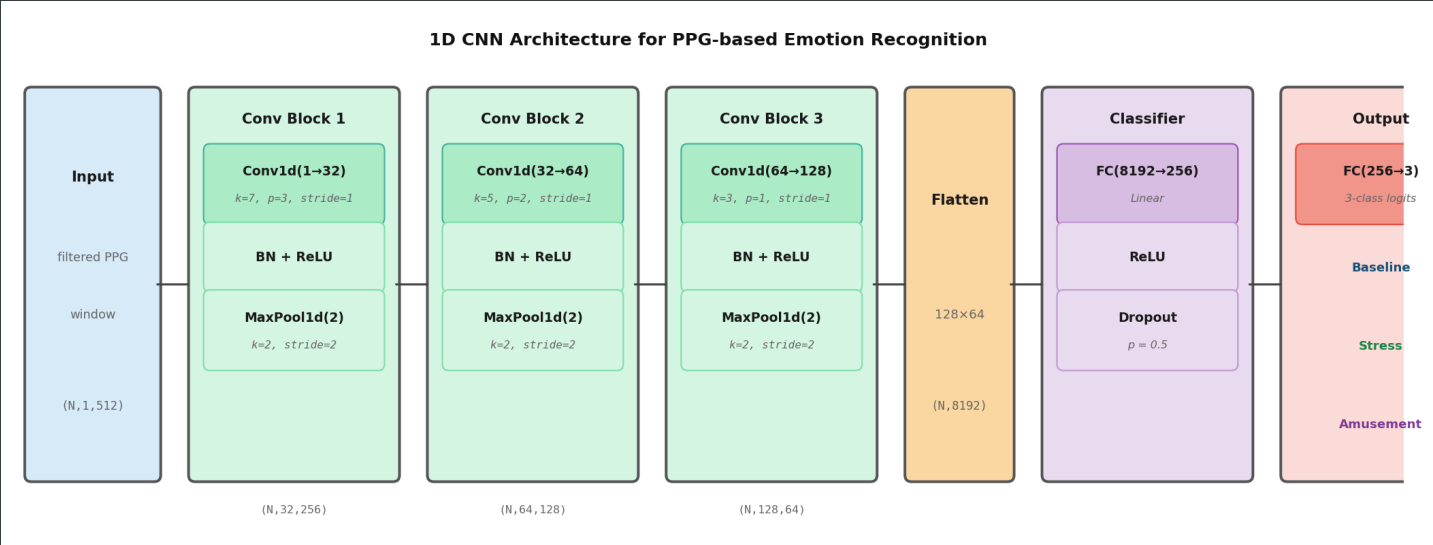


Figure 5a. *1D CNN architecture for PPG-based emotion recognition. The input is a filtered, per-segment z-score-normalized PPG window of shape (N, 1, 512). Three convolutional blocks progressively compress the temporal dimension while increasing channel depth from 1 to 32, 64, and 128. Each block applies Conv1d, BatchNorm, ReLU, and MaxPool1d(2), halving the sequence length at each stage. The resulting (N, 128, 64) feature map is flattened to 8192 dimensions, then passed through a two-layer classifier head: FC(8192→256) with ReLU and Dropout(0.5), followed by FC(256→3). The final layer produces raw logits for the three emotion classes (Baseline, Stress, and Amusement). No explicit Softmax is applied because CrossEntropyLoss incorporates log-softmax internally during training.

Decreasing kernel sizes allow early layers to capture long-range pulse morphology while later layers focus on local waveform features; Dropout(0.5) regularizes against overfitting on ~6,600 training samples.

The model was trained for 30 epochs using Adam ($\text{lr} = 1\text{e-}3$, $\text{weight_decay} = 1\text{e-}4$) with a StepLR scheduler ($\times 0.5$ every 10 epochs), batch size 64, and CrossEntropyLoss on labels remapped to 0/1/2. The training loss curve is shown in Figure 5.

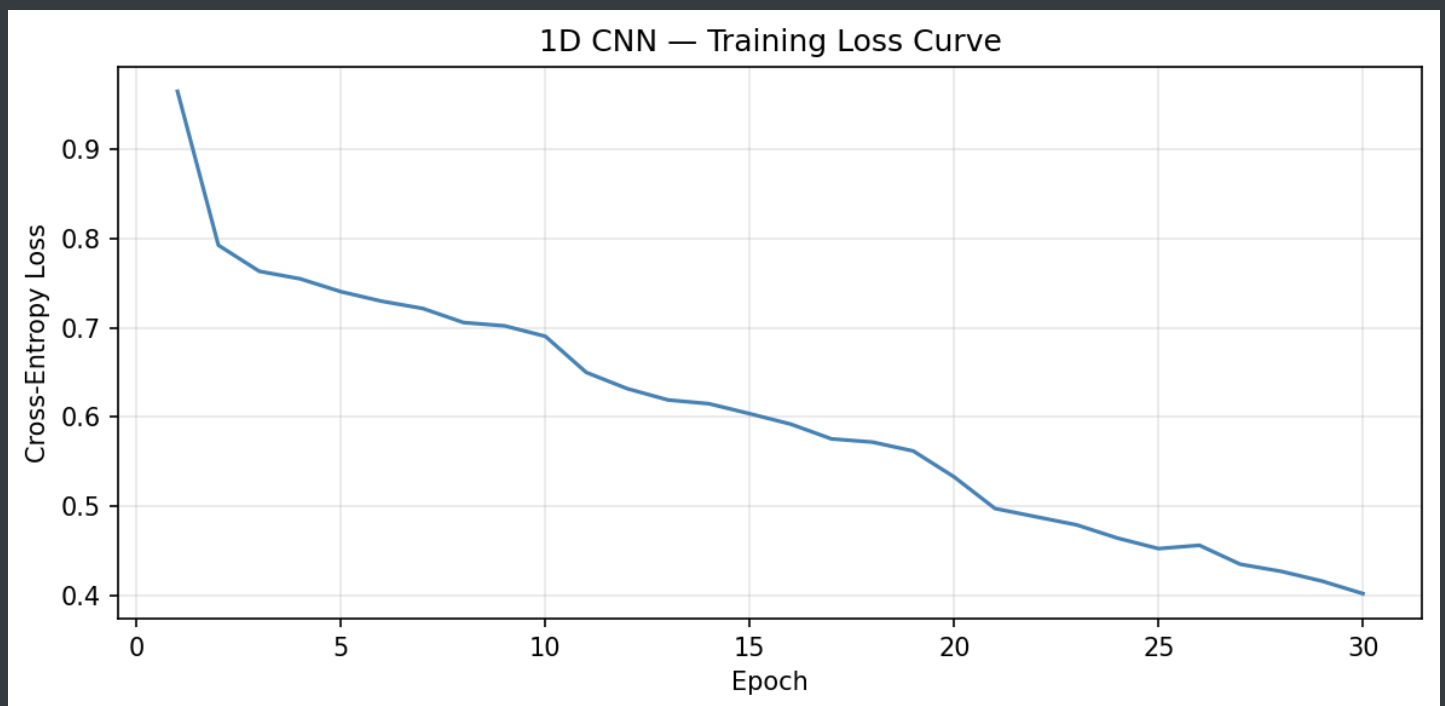


Figure 5. *Cross-entropy training loss per epoch. The StepLR scheduler produces visible loss drops at epochs 10 and 20. Loss decreases steadily from ~1.05 to 0.46, indicating consistent learning without divergence.*

Section 3 : Experiments & Analysis (10 pts)

3.1 Evaluation Protocol

All models are evaluated on subjects S15, S16, S17 (held out from all training). The test set contains 1,682 windows: Baseline 884 (52.6%), Stress 520 (30.9%), Amusement 278 (16.5%).

Because this is a three-class emotion recognition task rather than a binary detection problem, I report accuracy, macro-F1, per-class F1, per-class recall, and normalized confusion matrices, rather than binary-oriented metrics such as AUC-ROC or specificity. AUC-ROC and specificity are designed for binary outcomes; while they can be extended to multi-class settings via one-vs-rest averaging, that aggregation can obscure which specific classes are confused and by how much. Per-class recall defined as the fraction of true windows for each emotion state that the model correctly identifies, is particularly informative here because it directly exposes class-level detection gaps that aggregate accuracy masks. This is especially relevant for Amusement, where a model can achieve above-chance overall accuracy while completely failing to identify the class. Macro-F1 complements recall by also penalizing false positives, providing a balanced per-class view across the imbalanced test distribution.

3.2 Results

Table 1 : Model Comparison (Test Set: S15, S16, S17)

Model	Accuracy	Macro F1	F1 — Baseline	F1 — Stress	F1 — Amusement
Random Forest (tuned)	62.7%	56.4%	71.7%	61.1%	36.4%
Logistic Regression	57.8%	40.6%	67.8%	54.0%	0.0%
SVM (RBF)	62.4%	52.5%	71.6%	60.2%	25.8%
1D CNN	64.8%	55.6%	69.8%	71.5%	25.5%

Random chance baseline: 33.3% accuracy.

All models substantially outperform random chance. The CNN achieves the highest accuracy (64.8%) while Random Forest achieves the highest macro-F1 (56.4%). This divergence is driven by their contrasting per-class profiles discussed in Section 3.4.

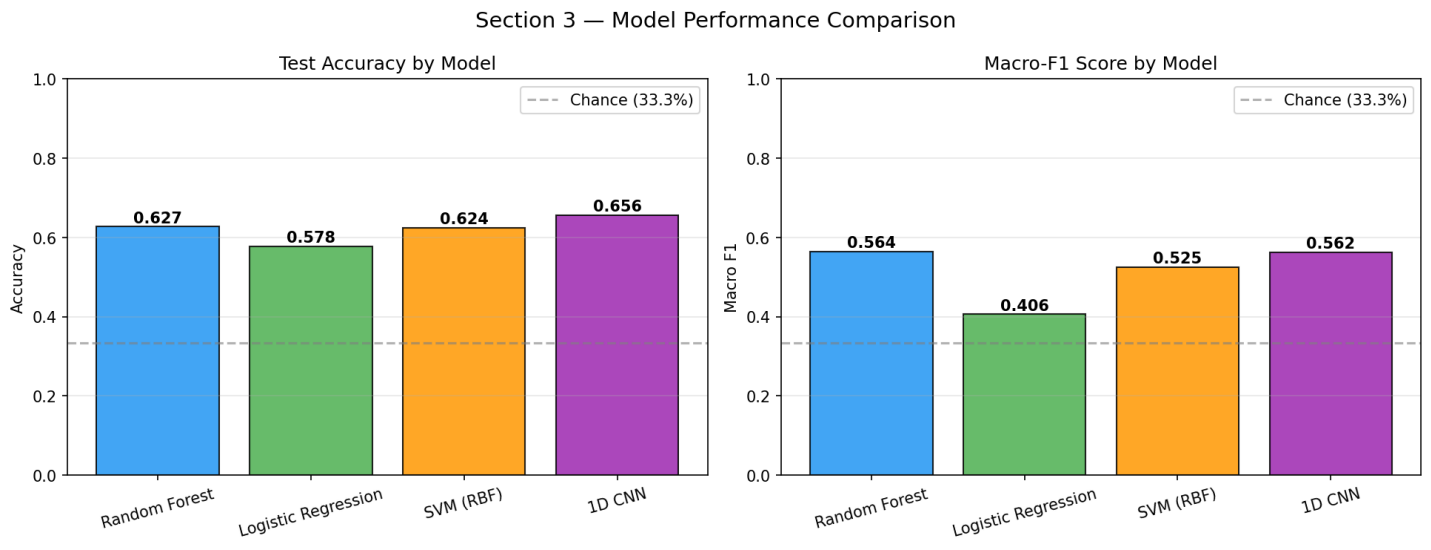


Figure 6. Accuracy (left) and macro-F1 (right) for all four models. The dashed line marks 33.3% chance. The CNN leads on accuracy; Random Forest leads on macro-F1.

3.3 Confusion Matrices

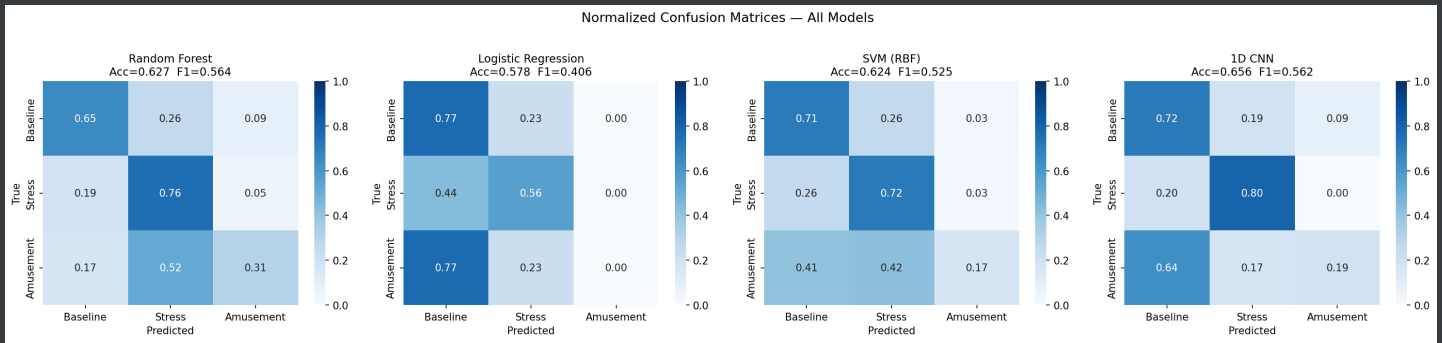


Figure 7. Row-normalized confusion matrices. Diagonal values are correct-classification rates; off-diagonal values are error rates. The pattern of Amusement being confused predominantly with Stress is broadly consistent with the confusion pattern reported in the original WESAD benchmark [1].

Table 2 : Per-Class Recall (Test Set: S15, S16, S17)

Model	Recall — Baseline	Recall — Stress	Recall — Amusement
Random Forest (tuned)	65.0%	76.0%	30.6%
Logistic Regression	77.0%	56.0%	0.0%
SVM (RBF)	71.3%	71.7%	16.9%
1D CNN	69.9%	80.4%	19.4%

Computed directly from the confusion matrices in Figure 7: each value is the CM diagonal element divided by the true class total (884 Baseline, 520 Stress, 278 Amusement windows).

The recall table reveals several consistent patterns. Stress recall is highest for the CNN (80.4%) and RF (76.0%), both substantially above Logistic Regression (56.0%). Amusement recall is poor across all models: RF achieves the best at 30.6%, while the CNN recovers only 19.4%. Most strikingly, Logistic Regression predicts zero Amusement windows correctly, classifying all 278 Amusement test samples as either Baseline (77.3%) or Stress (22.7%). All models misclassify Amusement primarily as Stress rather than as Baseline (RF: 52.2% of Amusement windows sent to Stress; SVM: 42.1%).

3.4 Per-Class F1 Comparison

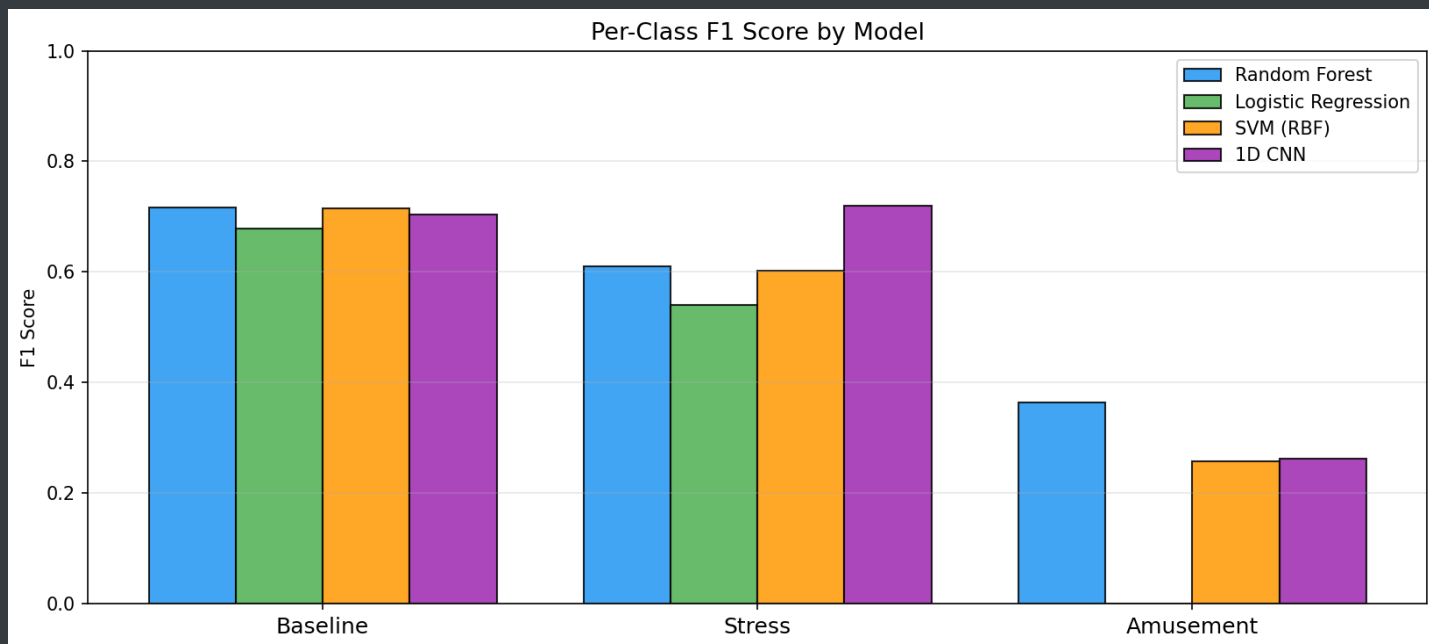


Figure 8. Per-class F1 scores across all four models. Baseline and Stress are reasonably well recognized; Amusement F1 is universally low, a pattern consistent with the original WESAD benchmark findings [1].

3.5 Analysis and Interpretation

Overall Performance

All models substantially outperform random chance (33.3%), confirming that wrist PPG contains meaningful discriminative information for three-class emotion recognition. My best result (CNN: 64.8% accuracy, 55.6% macro-F1) is consistent with the lower end of the wrist BVP single-modality range reported in the original WESAD paper; a precise numerical comparison is not straightforward, however, because my study uses shorter windows (8 s vs. 60 s), a held-out 3-subject split rather than full LOSO over 15 subjects, and a different classifier set [1]. The gap between wrist-only and chest-only results in the WESAD benchmark (~64–70% vs. ~80%) illustrates the ceiling that single-modality wrist BVP imposes on three-class performance, and motivates the multimodal extensions discussed under Limitations.

Why CNN Wins on Accuracy but not Macro-F1

The 1D CNN achieves the highest accuracy (64.8%) and the best Stress F1 (71.5%). Its Amusement F1 (25.5%), however, is comparable to SVM (25.8%) and worse than RF (36.4%). Random Forest distributes recognition ability more evenly across all three classes, yielding the highest macro-F1 (56.4%). The CNN's accuracy advantage is driven primarily by its dominant Stress performance; because Amusement recognition remains poor, its macro-F1 is pulled down relative to RF.

Baseline: Easiest Class

Baseline achieves the most consistent F1 across models (67.8%–71.7%). During the 20-minute resting period, parasympathetic dominance produces low, regular heart rate with high RMSSD and pNN50, a stable physiological fingerprint that all models recognize reliably. The long duration and absence of transient arousal events makes Baseline the least variable condition.

Stress: Well-Recognized by Non-Linear Models

Stress is well-classified by non-linear models (RF: 76.0% recall, CNN: 80.4%) but less so by Logistic Regression (56.0%). The TSST is specifically engineered to produce consistent, large-magnitude sympathetic responses: HR rises, RMSSD and SDNN fall sharply, and the LF/HF ratio increases. WESAD self-report data confirms this: STAI anxiety scores after TSST (18.5) were far higher than after Baseline (10.8) or Amusement (9.3) [1, Table 2]. These strong, consistent changes explain why non-linear classifiers exploit the Stress class effectively. The failure of Logistic Regression (56.0% recall) confirms that the Baseline/Stress boundary is non-linear in this 11-dimensional HRV feature space.

Amusement: Universally Difficult

Amusement is the hardest class for every model (F1: 0.0%–36.4%). This is consistent with the finding in the original WESAD paper, which noted that distinguishing Baseline from Amusement was difficult because the physiological changes elicited by the amusement condition were subtle, and self-reports indicated a smaller affective response compared to stress [1]. The quantitative self-report data bears this out: amusement STAI scores (9.3 ± 2.0) were nearly identical to baseline (10.8 ± 1.9), while arousal was only marginally elevated [1, Table 2]. Watching humorous videos produces mild positive arousal rather than sustained sympathetic

activation, so HR, RMSSD, and SDNN remain largely in the Baseline range. The counterintuitive Amusement-to-Stress confusion (RF: 52.2%, SVM: 42.1%) likely reflects transient HR accelerations during the funniest moments of clips, pushing those 8-second windows into the stress-like feature region.

Logistic Regression's complete failure on Amusement ($F1 = 0.0\%$) is the clearest illustration of why linear boundaries are insufficient: Amusement occupies a region overlapping both Baseline and Stress in every linear feature dimension, so no hyperplane can isolate it. The pattern is broadly consistent with the WESAD benchmark, where non-linear ensemble methods outperformed linear classifiers on the three-class problem [1].

Feature-Based vs. End-to-End

The CNN's superior Stress recall (80.4% vs RF's 76.0%) suggests that PPG waveform morphology carries discriminative Stress information beyond what the 11 HRV statistics capture, possibly changes in pulse shape, dicrotic notch prominence, or waveform asymmetry caused by increased peripheral vascular resistance under sympathetic activation. The CNN can learn these morphological signatures directly. However, the CNN shows no advantage for Amusement (19.4% recall vs RF's 30.6%), where handcrafted statistical and HRV features appear to provide better discriminative signal.

This contrast illustrates precisely the problem addressed by Rashid et al. [2]. Their H-CNN architecture concatenates handcrafted HRV-related features including the mean and standard deviation of HR and HRV, pNN50, LF/HF, and frequency-band energy components, with CNN-learned waveform representations before the final classification layer, operating exclusively on the wrist BVP signal from WESAD. Under LOSO evaluation with 60-second windows, the H-CNN achieved 75.21% accuracy and 64.15% macro-F1 for three-class classification, outperforming the best traditional classifier (LDA: 70.17% accuracy, 54.72% F1) by approximately 5% in accuracy and 10% in macro-F1, and outperforming the normal CNN (68.52% accuracy, 57.67% F1) by approximately 7% in both metrics. The result demonstrates that the two paradigms are genuinely complementary: domain-specific HRV features encode physiological knowledge that a CNN alone may not reliably learn from limited wrist data, while CNN-extracted morphological features capture signal patterns that handcrafted statistics cannot fully represent.

My current implementation does not reproduce this hybrid architecture. I train the CNN and the classical models as entirely separate pipelines; there is no feature-concatenation layer that jointly exploits both feature types. Implementing a true H-CNN-style hybrid by combining my 11 HRV features with CNN-learned representations would be a direct and well-motivated extension of this work, and based on the results in [2], is likely to improve performance on both Stress and the difficult Amusement class. Adopting longer windows (60 s) would also yield more reliable LF/HF estimates than my 8-second approach, though at the cost of substantially fewer training samples per subject.

Limitations

1. **LF/HF approximation:** 8-second windows yield 0.125 Hz spectral resolution, making LF/HF an approximation. Clinical standards recommend at least 5-minute windows [6]; both the original WESAD benchmark [1] and Rashid et al. [2] used 60-second windows.
2. **Evaluation variance:** Testing on only 3 subjects introduces high variance. WESAD subject-specific accuracies ranged from 69% to 98% [1], highlighting strong inter-subject variability that a 3-subject holdout cannot fully capture.
3. **Single modality:** The WESAD paper shows that incorporating additional wrist sensors alongside BVP substantially raises three-class accuracy [1]. Multimodal fusion is the most direct path to improving Amusement recognition.
4. **Class imbalance:** Baseline windows constitute 52.6% of the test set. Class-weighted training could further improve Amusement performance.

References

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Appendix — Code

File	Contents
Section1_DataProcessing.ipynb	Full Section 1 pipeline demonstrated on subject S2, including all visualizations
Section2_3_Classification.ipynb	Multi-subject processing, all four classifiers, and Section 3 comparative analysis